

Microeconomics: In-class Quiz 1

Fall 2026

Student ID:

Name:

Instructions

1. Do NOT flip over this page until every student receives this quiz. Your TA will let you know when you can start.
2. During this closed-book quiz, you cannot consult any materials.
3. If you are unable to explain your reasoning in English, it is okay to write in Korean.
4. Should you need more spaces, use the backside of the page, with clearly indicating the relevant quiz number.

Honor Code: Cheating on exams or quizzes, plagiarizing someone else's answers as one's own, or any other instance of academic dishonesty violates the standards of academic integrity.

Confidentiality Code: Sharing the information of the exam or quiz contents with other students in any form and medium is strongly prohibited, as it raises information inequity.

I, _____, consent to the Honor Code and the Confidentiality Code.
(write your name)

1. Suppose that the elasticity of demand for almonds with respect to the price of peanuts is -1.3 . Then, are almonds and peanuts substitutes?

2. In the market for coffee, suppose a severe frost destroys a large portion of the coffee bean crop. Simultaneously, a highly publicized medical study details new health benefits of drinking coffee. How will these events affect the equilibrium price and quantity of coffee?

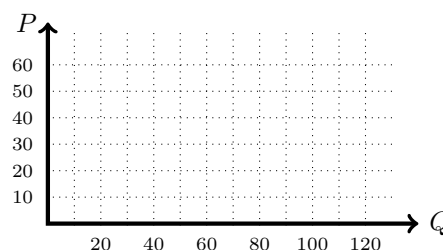
1. Both the equilibrium price and quantity will increase.
2. The equilibrium price will decrease, but the equilibrium quantity cannot be predicted.
3. The equilibrium price will increase, but the equilibrium quantity cannot be predicted.
4. The equilibrium quantity will decrease, but the effect on price cannot be predicted.

3. Suppose the government imposes a \$2 per unit tax on sellers of a good. The price elasticity of demand for this good is -0.4 , and the price elasticity of supply is 1.8 . Which of the following statements is true regarding the economic burden (incidence) of this tax?

1. Sellers will bear a larger share of the tax burden as they are responsible for the statutory taxes.
2. Buyers will bear a larger share of the tax burden as demand is more inelastic than supply.
3. Sellers will bear a larger share of the tax burden as supply is more elastic than demand.
4. The tax burden will be shared equally between buyers and sellers.

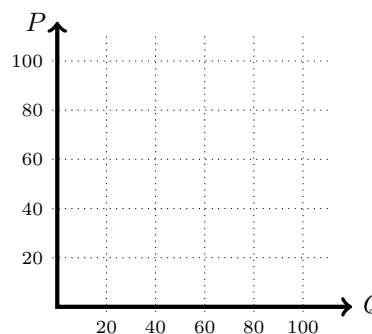
4. In the market for widgets, the quantity demanded and quantity supplied are expressed as $Q^D = 120 - 2P$ and $Q^S = 30 + P$, where P is the price per widget and Q measures the number of widgets. Suppose the government sets a price ceiling of \$20 per widget.

- (a) Draw the inverse supply and demand functions on the graph on the right.
- (b) Find the equilibrium quantity without the price ceiling.
- (c) Find the equilibrium quantity with the price ceiling.
- (d) Calculate the consumer surplus with the price ceiling.



5. The market for textbooks is perfectly competitive. The market demand is given by $Q^D = 100 - P$, and the market supply is given by $Q^S = P - 20$, where P is the price in dollars and Q is the quantity of textbooks.

- (a) Find the equilibrium price and quantity in the absence of any government intervention.
- (b) Suppose the government imposes a specific tax of \$10 per textbook on sellers. Find the new equilibrium quantity.
- (c) Calculate the price paid by buyers and the price received by sellers after the tax is imposed.
- (d) Calculate the deadweight loss (DWL) caused by this tax.



6. The table below shows the annual expenditure data of an office worker before and after a salary increase. Assume that the prices of all goods and services remained constant between 2025 and 2026.

Year	Income	Groceries	Beverages	Telecom	Overseas Travel	Home Appliances
2025	\$50,000	\$1,000	\$100	\$400	\$20,000	\$7,000
2026	\$60,000	\$1,200	\$130	\$500	\$23,000	\$7,100

Which of the following statements is **correct** regarding this worker's consumption patterns?

1. Overseas travel is a luxury good because the expenditure increased significantly by \$3,000.
2. Beverages are considered a necessity because the absolute increase in expenditure is very small.
3. Based on the income elasticity of demand, telecom services are considered a luxury good.
4. The income elasticity of demand for home appliances is negative, meaning it is an inferior good.